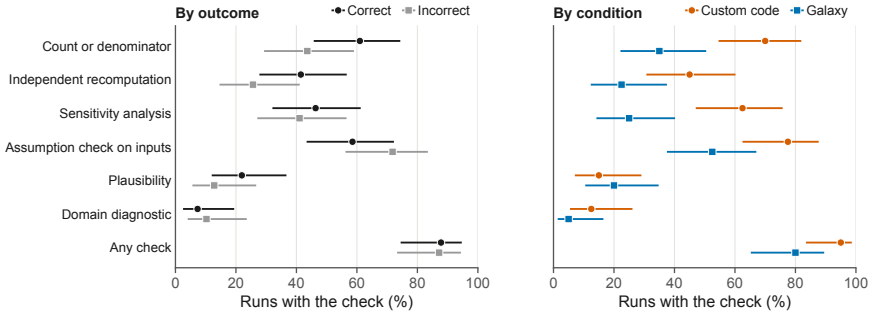
a Verification checks in 80 coded runs (coder blind to the grade)

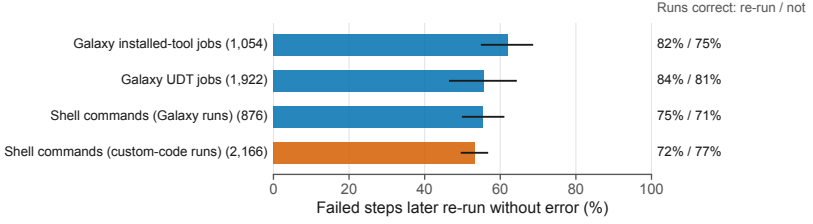
10 runs per benchmark × condition × outcome; 95% Wilson intervals



A check changed the method or answer in
6 of 41 correct runs
11 of 39 incorrect runs

b Failed steps later re-run without error in the same run

Galaxy jobs by tool; shell commands by the analysis program or script they ran (inline code excluded)



c A parameter check that stopped a job

Task	bix-43-q4, GPT-5.5, Galaxy, replicate 1 (Reactome enrichment)
Tool	gseapy_enrichr 0.1.0+galaxy2
First request	nested: gene_sets = {source, library_name, organism}
Interface check	gene_sets|library_name: requested Reactome_2022, Galaxy would bind (empty); the job was not submitted
Agent response	Resubmitted with flat parameter keys
Second request	flat: gene_sets|library_name = Reactome_2022; parameters matched; job ran

d A selected case: variant-status-q1 (CompBioBench)

Read-end artefacts mimic an alternate allele; answers are not shown

	Custom code	Galaxy
GPT-5.5	● ○ ○	○ ○ ○
GPT-5.6 Sol	○ ○ ○	○ ● ●
GPT-5.6 Luna	○ ○ ○	○ ○ ○
DeepSeek V4 Pro	○ ● ○	○ ○ ○

● accepted ○ rejected □ ran a read-position diagnostic (5 runs)
4 of 24 runs accepted. GPT-5.6 Sol in Galaxy ran the diagnostic as a UDT in replicates 2 and 3, and both were accepted; 3 custom-code runs also ran it (1 accepted).